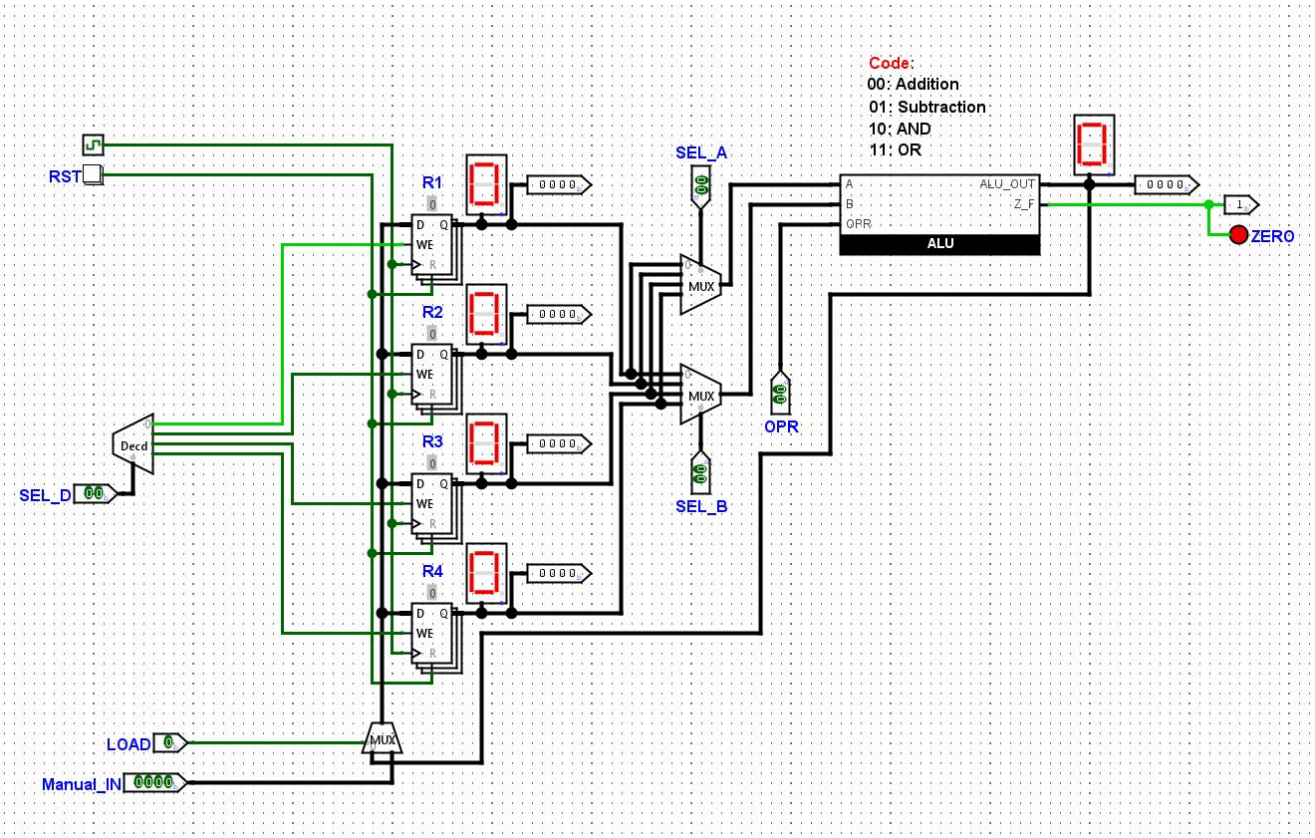
4-BIT MICROPROCESSOR DATAPATH

AN RTL-STYLE LEARNING PLATFORM

A four-register datapath with a custom 4-function arithmetic logic unit, independent operand selection, decoder-controlled register write-back, and a combinational zero flag.

4 × 4	4	2 × 4:1	ZF
BIT REGISTERS	ALU OPERATIONS	SOURCE MUXES	ZERO FLAG

ACTUAL LOGISIM-EVOLUTION SCHEMATIC



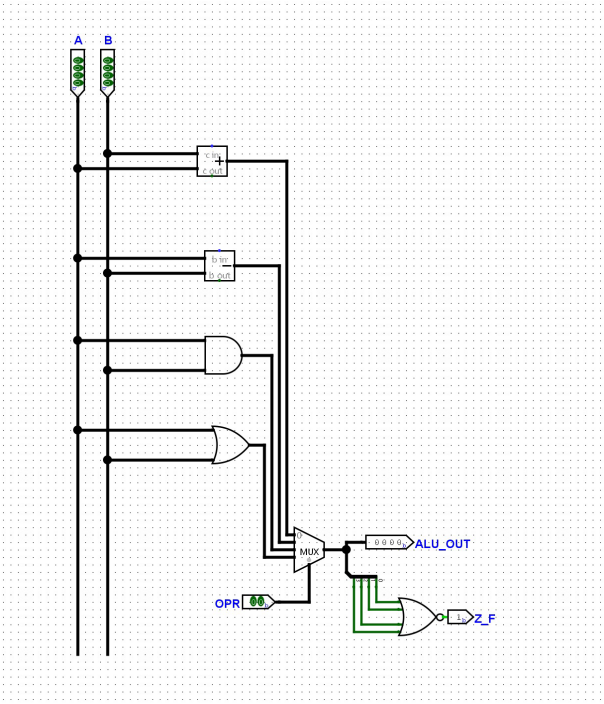
EXECUTION FLOW

Initialize registers → select two source operands → compute with the ALU → write back to the decoder-selected destination on a clock edge.

4-BIT MICROPROCESSOR DATAPATH

FOUR OPERATIONS, ONE RESULT BUS

The custom ALU combines a 4-bit adder, subtractor, bitwise AND/OR paths, an operation multiplexer, and four-bit NOR zero detection.



OPERATION CONTROL

00	ADD	(A + B) & 0xF
01	SUB	(A - B) & 0xF
10	AND	A & B
11	OR	A | B

STATUS FLAG

ZF = 1 if and only if the 4-bit ALU output is 0000. The zero detector is implemented with a four-input NOR gate.

SELECTED SIMULATION EXAMPLES

ADD

5 + 3

RESULT: 8

ZF = 0

SUB

4 - 4

RESULT: 0

ZF = 1

REGISTER LOAD

one selected register

RESULT: updated

others hold